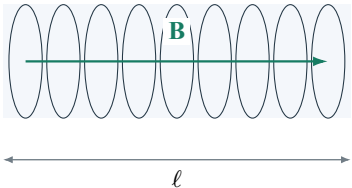
M7-002　长螺线管的磁场能

N 匝，横截面 A



I：绕组电流

$$B = \mu N I/\ell$$

$$L = \mu N^2 A/\ell, \quad U_B = \tfrac{1}{2} L I^2$$

长螺线管近似；内部积分体积 $A\ell$。